

Optimize Power BI Models: Master Tools & Techniques for Peak Performance

Develop data models will perform





ONE DAY



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Principal Solution Architect

One Day Business Intelligence



- ✓ 10+ Years of Power BI Experience
- ✓ 9+ Year of Training Experience
- ✓ Microsoft Certified Solution Expert – Data Management and Analytics
- ✓ Microsoft Certified Trainer
- ✓ Azure Enterprise Data Analyst
- ✓ Fasttrack Recognized Solution Architect



Adoption is everything!

70-85% of data & analytics projects fail to meet their objectives. The root cause is rarely technical - it's adoption.

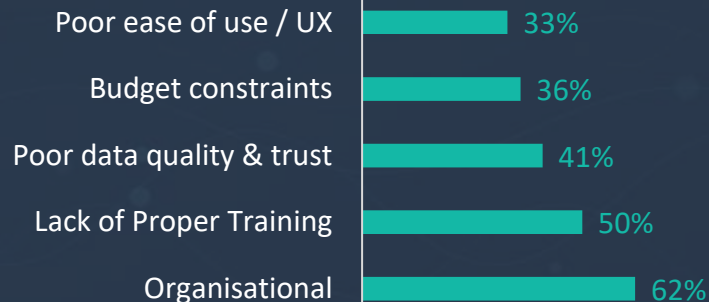


Why Delivered Data Projects Fail to Get Adopted

Adoption Reality

- BI adoption stuck at ~25% for 7+ years
- only 29% of employees actively use analytics tools
- \$2.3T lost per year to failed digital transformation globally

Top Barriers





What determines Adoption?

Delivered Value

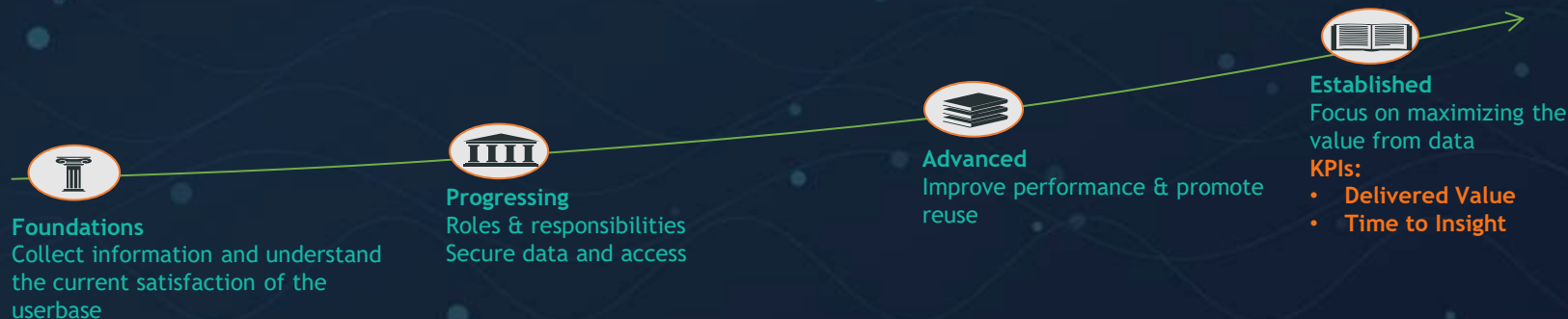
- Executive Sponsorship
- Data Quality First
- Self-Service Tooling
- Integration & Governance
- Training & Support

Time to Insight

- **0.1 seconds** is the limit for the system feeling instantaneous — no feedback needed beyond showing the result
- **1.0 second** is the limit for the user's flow of thought to stay uninterrupted — the user notices the delay but doesn't need special feedback
- **10 seconds** is the limit for keeping the user's attention focused on the dialogue

Jakob Nielsen – Usability Engineering (1993)

Governance Maturity Curve





Performance Analysis Tools

What is in the toolbelt?



Performance Analysis

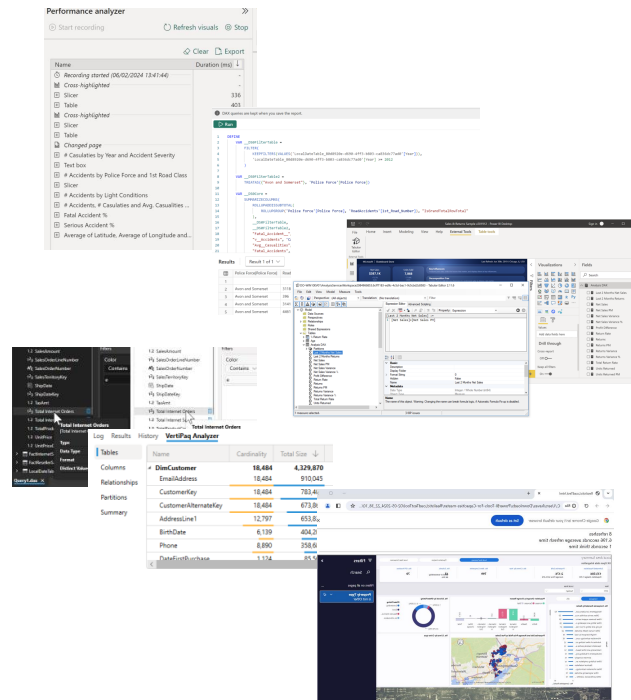
To provide good user experience and support adoption the performance of the report plays a crucial role.

The perceived performance are influenced by a lot of different factors:

- End user hardware
- Quality and speed of the network
- Security settings
- Physical distance from the server
- Complexity of the report
- **Quality of the model**

Tools of performance analysis

- Power BI Desktop
 - Performance Analyzer
 - Dax Query View
- Dax Studio
 - Query Traces
 - Vertipaq Analyzer
- Tabular Editor
 - Best Practice Analyzer
- SQL Server Profiler
- Azure Log Analytics
- Fabric Notebooks
 - Semantic Link
 - Model health
- Power BI Realistic Load Testing Tool





Built-in Tools - Power BI Desktop

First step in performance analysis

- Performance Analyzer
- Dax Query View
- Optional: Query Diagnostic

Performance analyzer

Start recording Refresh visuals Stop

Clear Export

Name	Duration (ms)
Recording started (06/02/2024 13:41:44)	-
Cross-highlighted	-
Slicer	336
Table	402
Cross-highlighted	-
Slicer	-
Table	-
Changed page	-
# Casualties by Year and Accidents	-
Text box	-
# Accidents by Police Force and Area	-
Slicer	-
# Accidents by Light Conditions	-
# Accidents, # Casualties and Area	-
Fatal Accident %	-
Serious Accident %	-
Average of Latitude, Average of Longitude	-

DAX queries are kept when you save the report.

Run

```
1 DEFINE
2 VAR __DSRFilterTable =
3     FILTER(
4         KEEPFILTERS(VALUES('LocalDateTable_808920e-d690-4ff3-b803-ca836dc77ad0'[Year])),
5         'LocalDateTable_808920e-d690-4ff3-b803-ca836dc77ad0'[Year] >= 2012
6     )
7
8 VAR __DSRFilterTable2 =
9     TREATAS(("Avon and Somerset"), 'Police Force'[Police Force])
10
11 VAR __DSRCore =
12     SUMMARIZECOLUMNS(
13         ROLLUPADDSORTTOTAL(
14             ROLLUPGROUP('Police Force'[Police Force], 'RoadAccidents'[1st_Road_Number]), "IsGrandTotalRowTotal"
15         ),
16         __DSRFilterTable,
17         __DSRFilterTable2,
18         "Fatal_Accident_%", "Calculation"[Fatal Accident %],
19         "v_Accidents", "Calculation"[# Accidents],
20         "Avg_Casualties", "Calculation"[Avg. Casualties],
21         "Fatal_Accidents", "Calculation"[Fatal Accidents]
```

Results Result 1 of 1 Copy

	Police Force[Police Force]	RoadAccidents[1st_Roa...	[IsGrandTotalRowTotal]	[Fatal_Accident_...]	[v_Accidents]	[Avg
1				0.02	9962	
2	Avon and Somerset	3118		0.5	2	
3	Avon and Somerset	396		0.18	11	
4	Avon and Somerset	3141		0.17	6	
5	Avon and Somerset	4461		0.13	8	



Dax Studio

The ultimate tool for working with DAX queries.

- Robust Query builder
- DAX Profiler
- Query Performance Analyzer
- Access to DMV
- Analyze models – Vertipaq Analyzer
- Data Export capabilities

The screenshot displays the Dax Studio interface. On the left, the query editor shows a DAX query for 'Total Internet Orders'. On the right, the 'Vertipaq Analyzer' tool is open, displaying a table of model metadata.

Name	Cardinality	Total Size ↓
DimCustomer	18,484	4,329,870
EmailAddress	18,484	910,045
CustomerKey	18,484	783,400
CustomerAlternateKey	18,484	673,863
AddressLine1	12,797	653,839
BirthDate	6,139	404,208
Phone	8,890	358,687
DateFirstPurchase	1,174	85,568

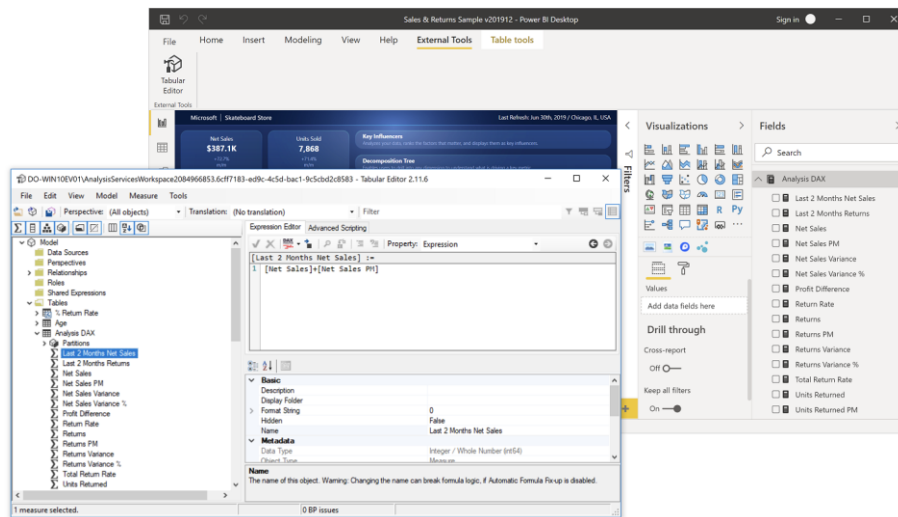
<https://daxstudio.org/>



Tabular Editor

Light weight model development tool.

- Fast – because it only works with metadata
- Efficient – C# scripting
- Integrate - CLI
- Best Practice Analyser
- Can be combined with Vertipaq information for complete model documentation



<https://tabulareditor.github.io/>

<https://www.elegantbi.com/post/vpaxtotabulareditor>

<https://github.com/microsoft/Analysis-Services/tree/master/BestPracticeRules>



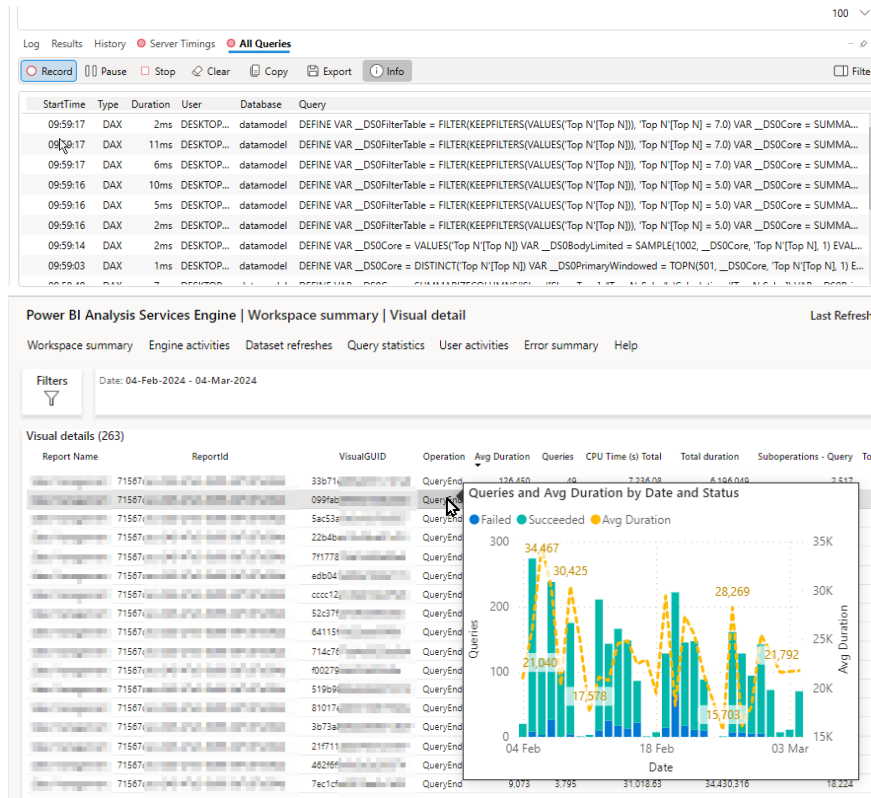
SQL Server Profiler & Azure Log Analytics

Profiler:

- Temporary monitoring
- Identification of performance problems

Log Analytics:

- Can be set up on the Workspace level
- Premium required
- Keeps the data for 30 days by default
- Relatively low cost
- [PBI Template is available](#)
- Combining with Power BI Audit information can enrich the analysis



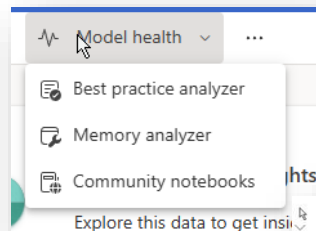
<https://learn.microsoft.com/en-us/power-bi/transform-model/log-analytics/desktop-log-analytics-overview>



Fabric Notebooks

Fabric enables automated programmatic analysis of multiple models.

- Notebooks - Semantic Link
- Model health
 - Best Practice analyzer
 - Memory analyzer



Run the Best Practice Analyzer on your semantic model

```
1 fabric.run_model_bpa(dataset=dataset, workspace=workspace)
```

[2] ✓ 7 min 10 sec - Command executed in 7 min 10 sec 379 ms by Mihaly Kawaii on 04/02/2026, 10:04:02

DAX Expressions	Error Prevention	Formatting	Maintenance	Performance
✗ (72) + ⚠ (14)	⚠ (1)	⚠ (67) + ⚠ (53)	⚠ (143) + ⚠ (4)	⚠ (143)
Rule Name				
Object Type		Object Name		Severity
Column references should be fully qualified		DateTableTemplate_cd14fca4-29d8-4938-973b-7dc264bda5c3[Day]		✗
Column references should be fully qualified		DateTableTemplate_cd14fca4-29d8-4938-973b-7dc264bda5c3[MonthNo]		✗
Column references should be fully qualified		DateTableTemplate_cd14fca4-29d8-4938-973b-7dc264bda5c3[Month]		✗
Column references should be fully qualified		DateTableTemplate_cd14fca4-29d8-4938-973b-7dc264bda5c3[QuarterNo]		✗
Column references should be fully qualified		DateTableTemplate_cd14fca4-29d8-4938-973b-7dc264bda5c3[Quarter]		✗
Column references should be fully qualified		DateTableTemplate_cd14fca4-29d8-4938-973b-7dc264bda5c3[Year]		✗
Column references should be fully qualified		LocalDateTable_04fe26d8-f2e5-4098-aa36-95d6aaf2b621[Day]		✗
Column references should be fully qualified		LocalDateTable_04fe26d8-f2e5-4098-aa36-95d6aaf2b621[MonthNo]		✗

Run the Memory Analyzer on your semantic model

```
1 fabric.model_memory_analyzer(dataset=dataset, workspace=workspace)
```

[4] ✓ 1 min 2 sec - Command executed in 1 min 2 sec 296 ms by Mihaly Kawaii on 04/02/2026, 09:57:59

Model Summary												
Tables												
Partitions												
Columns (Total Size)												
Columns (Temperature)												
Relationships												
Hierarchies												
Tables												
Table Name												
Type												
Row Count												
Total Size												
Dictionary Size												
Data Size												
Hierarchy Size												
Relationship Size												
User Hierarchy Size												
Partitions												
Columns												
% DB												
0	Company Data	Table	5,005,146	753,738,823	495,403,563	142,695,776	113,811,096	3,826,338	0	1	39	46.13%
1	Prod Code	Table	2,399,368	212,584,144	141,059,544	27,170,938	44,368,416	85,236	0	1	12	13.58%
2	Crime Data	Table	10,826,153	197,317,728	57,343,480	127,367,064	12,339,712	67,456	0	1	11	12.60%
3	Housing Market Data	Table	10,431,664	186,629,518	64,047,630	101,196,400	16,396,840	4,786,640	0	1	9	11.60%
4	LocalDateTable_345ed6b6-1550-4f72-ac7e-73c78d547819	Calculated Table	2,934,724	178,964,712	91,776,672	13,604,024	23,342,896	0	48,141,120	1	7	11.37%
5	GP Practice Data	Table	1,727,664	8,000,164	1,237,948	6,370,136	326,176	65,904	0	1	4	0.51%
6	Population Estimates	Table	1,737,650	5,361,777	1,919,409	3,979,728	293,120	69,520	0	1	5	0.34%
7	Schools	Table	24,037	4,806,240	3,741,120	299,008	672,168	93,844	0	1	22	0.31%
8	Population Trend Data	Table	1,379,560	4,635,304	212,120	4,377,792	63,152	440	0	1	6	0.30%
9	LocalDateTable_2c96dc8b-1766-4b15-890b-0c2a9f18a15a	Calculated Table	54,422	4,244,888	2,677,536	218,664	437,382	0	911,296	1	7	0.27%
10	Lower Super Output Area	Table	34,753	3,138,193	2,203,505	231,680	702,616	392	0	1	7	0.20%



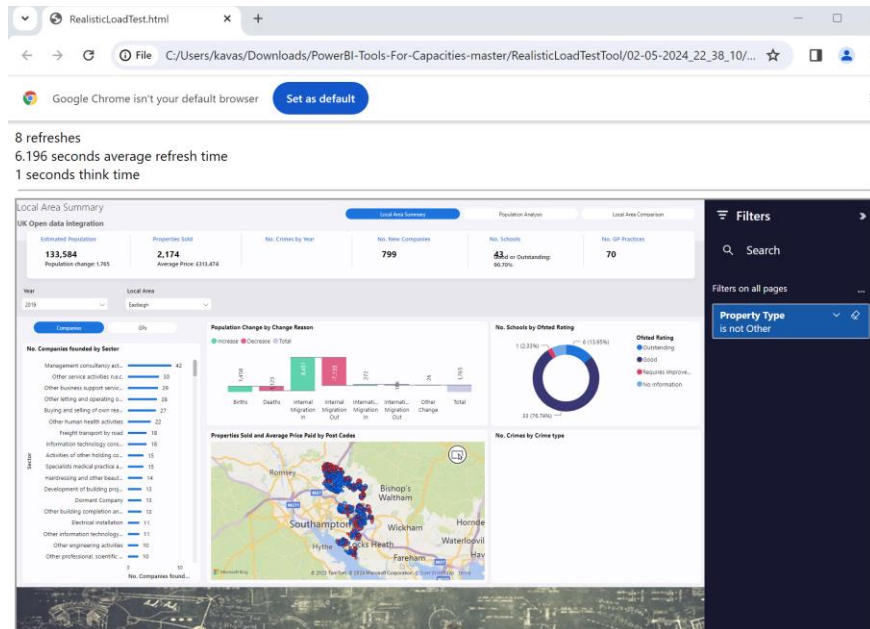
Realistic Load Testing Tool

End user experience testing tool

- Multi users experience (testing performance thresholds)
- Dynamic interactions with the reports
- Can be used to analyse the end-2-end performance

Important:

- Results should be treated as guidance not exact values
- Need to make sure that the machine running the tool is not the bottleneck -> Running Performance Monitor is recommended (perfmon)



<https://github.com/microsoft/PowerBI-Tools-For-Capacities/tree/master>

You can find me:

- [Website: www.onedaybi.com](http://www.onedaybi.com)
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- [Sessionize: https://sessionize.com/mihalykavasi/](https://sessionize.com/mihalykavasi/)
- [GitHub: https://github.com/KavasiMihaly/](https://github.com/KavasiMihaly/)
- [My Tabular Editor Scripts: https://github.com/KavasiMihaly/SelfServiceBIBlog/tree/main/TabularEditor](https://github.com/KavasiMihaly/SelfServiceBIBlog/tree/main/TabularEditor)

